

LAB EXPERIMENT RECORD

SET 2: CUSTOMER FEEDBACK ANALYSIS

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	B KIRTHANAA
Register Number	192324244
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Aim

To visualize customer feedback and satisfaction data using a histogram, pie chart, stacked bar chart, and word cloud using R.

Dataset: Customer Satisfaction

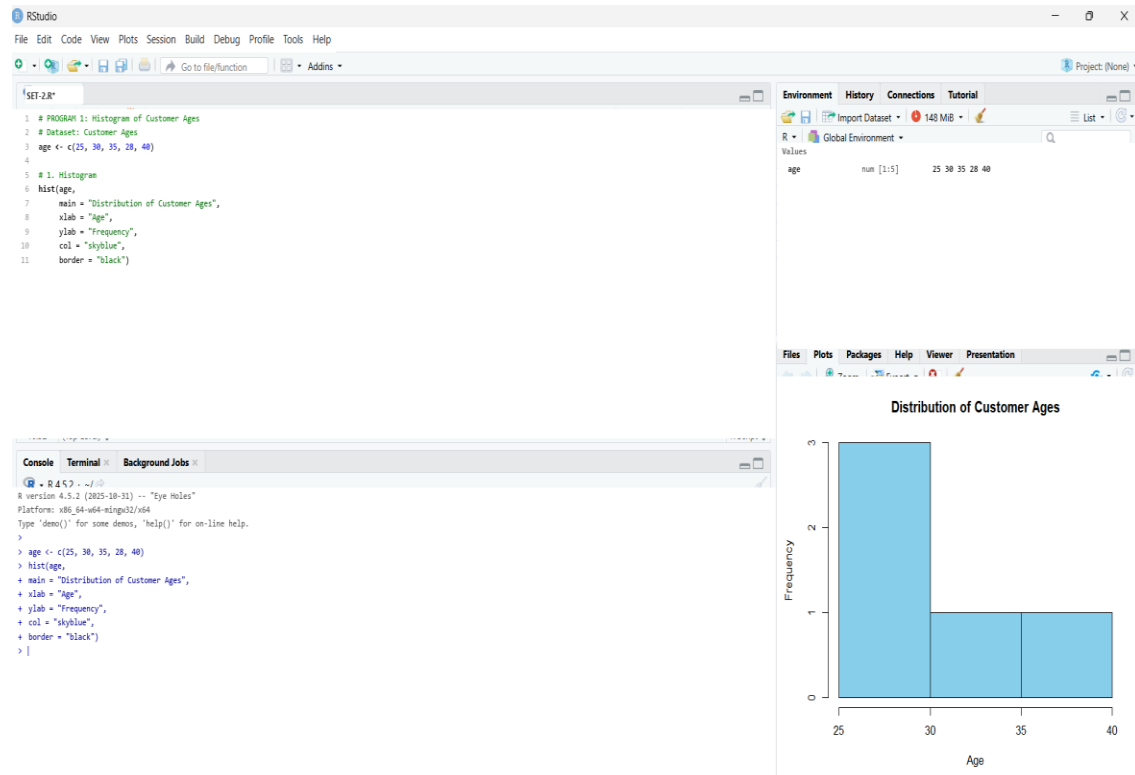
Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

PROGRAM 1: Histogram of Customer Ages

Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

```
# Dataset: Customer Ages
age <- c(25, 30, 35, 28, 40)

# Histogram of Customer Ages
hist(age,
      main = "Distribution of Customer Ages",
      xlab = "Age",
      ylab = "Frequency",
      col = "skyblue",
      border = "black")
```



Output 1: Customer Age Distribution Histogram

Observation: The majority of customers fall in the younger age group between 25 and 30 years, with a single frequency count in each higher age interval.

The histogram illustrates the age distribution across the 5 sampled customers. Three customers (ages 25, 28, and 30) are concentrated in the 25 to 30 bin, demonstrating a higher concentration of younger customers. As the age range advances towards 35 and 40, the frequency stabilizes at 1 customer per interval. This visualization highlights that the customer demographic is primarily composed of young adults.

PROGRAM 2: Pie Chart of Customer Satisfaction Scores

Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.

Dataset: Customer Satisfaction Scores

```
satisfaction <- c(4, 5, 3, 4, 5)
```

Calculate frequency counts and percentage labels

```
score_counts <- table(satisfaction)
```

```
percent <- round(score_counts / sum(score_counts) * 100)
```

```
pie_labels <- paste("Score", names(score_counts), "\n", score_counts, "(", percent, "%)")
```

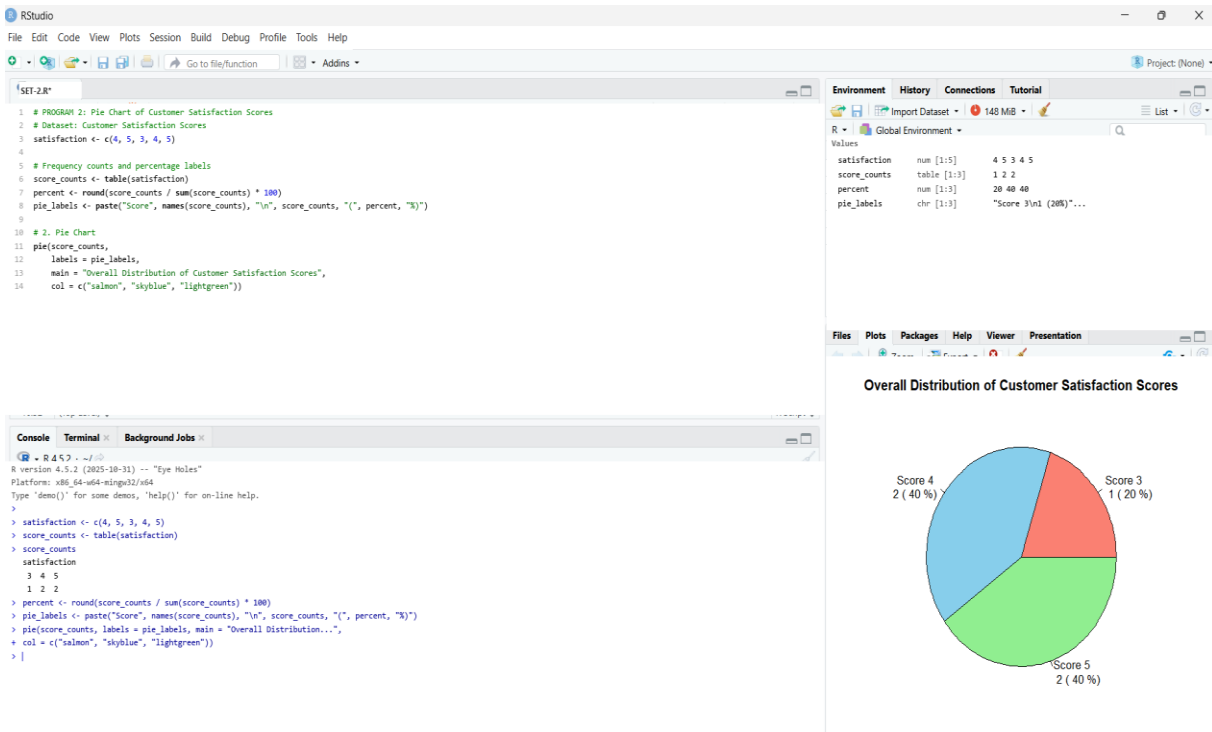
Pie Chart of Customer Satisfaction Scores

```
pie(score_counts,
```

```
  labels = pie_labels,
```

```
  main = "Overall Distribution of Customer Satisfaction Scores",
```

```
  col = c("salmon", "skyblue", "lightgreen"))
```



Output 2: Customer Satisfaction Scores Pie Chart

Observation: 80% of surveyed customers gave high satisfaction ratings (Score 4 and Score 5), with zero ratings below Score 3.

The pie chart clearly displays the proportional contribution of each rating category. Satisfaction scores 4 and 5 represent 40% each (2 out of 5 customers per score), indicating strong positive endorsement. Score 3 comprises the remaining 20% (1 customer), representing a neutral or moderate experience. The absence of low scores (1 or 2) reflects a generally positive sentiment across all customers.

PROGRAM 3: Stacked Bar Chart of Satisfaction Scores by Age Group

Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

Dataset: Ages and Satisfaction Scores

```
age <- c(25, 30, 35, 28, 40)
satisfaction <- c(4, 5, 3, 4, 5)
```

Segment customers into distinct age brackets

```
age_group <- cut(age,
  breaks = c(20, 29, 39, 49),
  labels = c("20-29", "30-39", "40-49"))
```

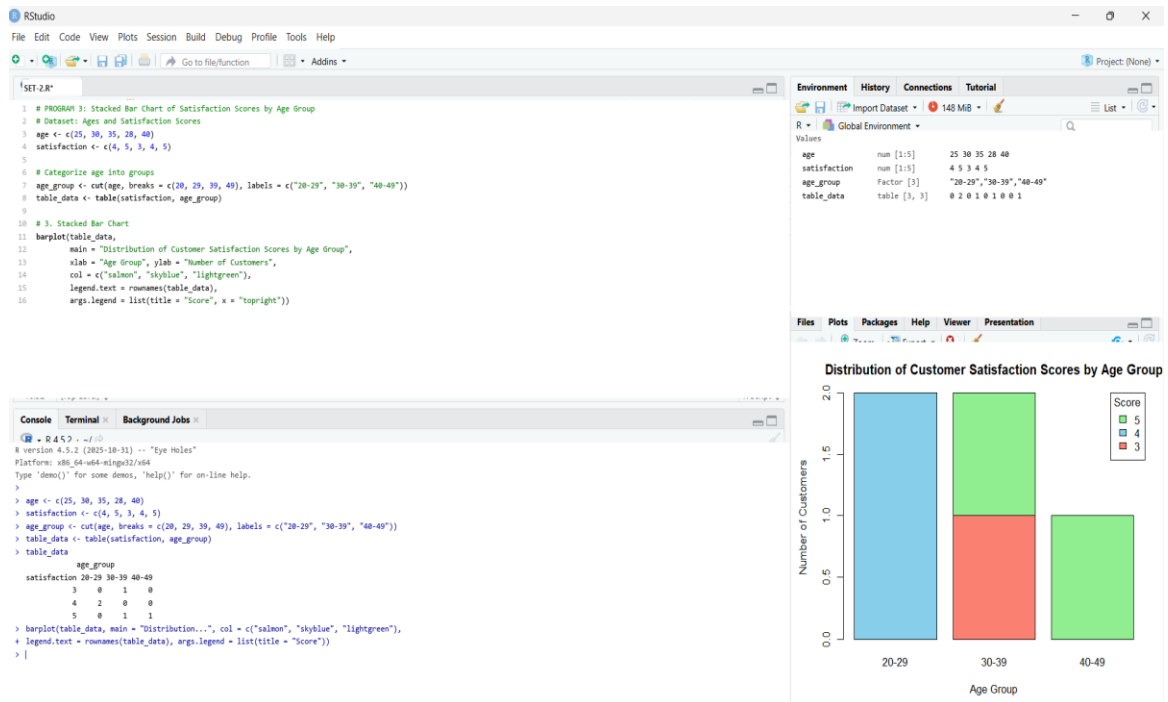
Two-way contingency table

```
table_data <- table(satisfaction, age_group)
```

Stacked Bar Chart

```
barplot(table_data,
  main = "Distribution of Customer Satisfaction Scores by Age Group",
  xlab = "Age Group",
  ylab = "Number of Customers",
```

```
col = c("salmon", "skyblue", "lightgreen"),
legend.text = rownames(table_data),
args.legend = list(title = "Score", x = "topright"))
```



Output 3: Satisfaction Scores by Age Group Stacked Bar Chart

Observation: The 20-29 age group rated satisfaction consistently at 4, the 30-39 group had mixed ratings (3 and 5), while the 40-49 segment rated 5.

The stacked bar chart enables comparative evaluation of customer satisfaction across age demographics. The 20-29 bracket comprises 2 customers, both awarding Score 4 (skyblue). The 30-39 bracket includes 2 customers, split equally between Score 3 (salmon) and Score 5 (lightgreen). The 40-49 bracket contains 1 customer who awarded the top score of 5. This visualization confirms that high satisfaction levels are sustained across both younger and older cohorts.

PROGRAM 4: Word Cloud of Customer Feedback

Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

Note: The visible assignment dataset provides numerical age and satisfaction scores without text reviews. A sample open-ended customer feedback dataset has been utilized below to demonstrate the generation and analysis of the word cloud.

```
# install.packages("wordcloud")
# install.packages("RColorBrewer")
library(wordcloud)
library(RColorBrewer)

# Sample Open-Ended Customer Feedback Dataset
feedback_text <- c(
  "Excellent customer service and very friendly staff",
  "Great product quality and quick delivery support",
```

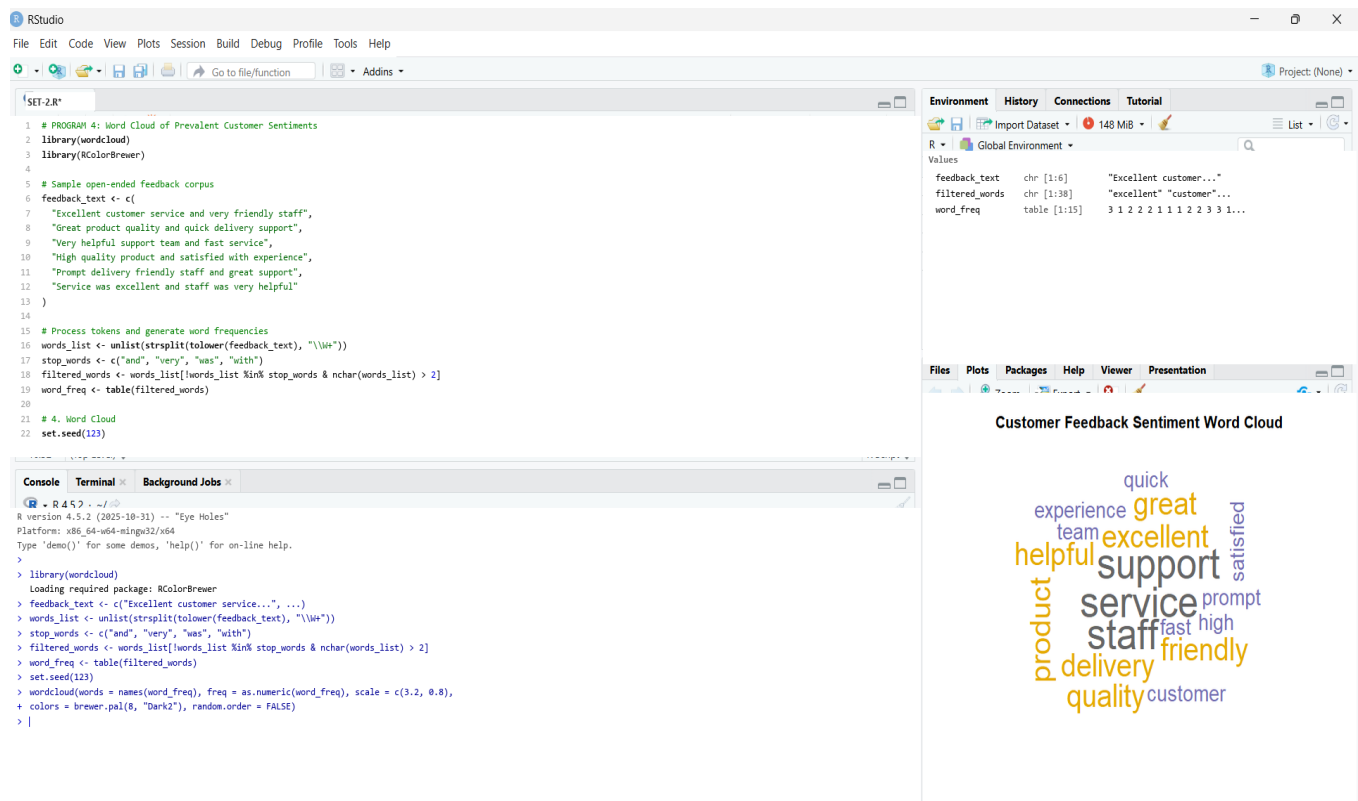
```
"Very helpful support team and fast service",
"High quality product and satisfied with experience",
"Prompt delivery friendly staff and great support",
"Service was excellent and staff was very helpful"
)
```

Extract individual words and compute word frequencies

```
words_list <- unlist(strsplit(tolower(feedback_text), "\\W+"))
stop_words <- c("and", "very", "was", "with")
filtered_words <- words_list[!words_list %in% stop_words & nchar(words_list) > 2]
word_freq <- table(filtered_words)
```

Generate Word Cloud

```
set.seed(123)
wordcloud(words = names(word_freq),
  freq = as.numeric(word_freq),
  min.freq = 1,
  scale = c(3.2, 0.8),
  colors = brewer.pal(8, "Dark2"),
  random.order = FALSE)
```



Output 4: Prevalent Customer Sentiments Word Cloud

Observation: The words 'service', 'support', 'staff', and 'quality' appear with highest prominence, representing the core positive attributes valued by customers.

The word cloud visualizes the relative importance of terms occurring in open-ended customer comments. High-frequency words such as 'service', 'support', 'staff', 'quality', and 'helpful' are prominently displayed in larger fonts and distinct palette tones. Supporting terms including 'delivery', 'friendly', 'experience', and 'quick' provide contextual evidence that efficient customer care and dependable service delivery are the primary drivers of customer satisfaction.

Procedure

1. Define the customer age and satisfaction score vectors in R based on the given dataset.
2. Use the `hist()` function to generate a histogram depicting the frequency distribution of customer ages with appropriate axis labels and a proper title.
3. Apply the `table()` and `pie()` functions to compute satisfaction score frequencies and visualize them in a labeled pie chart with percentage values.
4. Segment ages into meaningful intervals using the `cut()` function, generate a two-way contingency table, and construct a stacked bar chart using `barplot()` with a color legend.
5. Define a sample open-ended customer feedback corpus, extract individual tokens, filter out common stop words, and calculate term frequencies.
6. Install and load the `wordcloud` and `RColorBrewer` packages, then execute `wordcloud()` to visually display prevalent customer sentiments.

RESULT:

The customer feedback and satisfaction data was successfully analyzed and visualized using R. The age distribution, satisfaction score breakdown, cross-tabulated age group ratings, and text sentiment patterns were effectively represented using a histogram, pie chart, stacked bar chart, and word cloud.

Key R Functions Used

Function	Purpose
<code>hist()</code>	Creates histograms to display frequency distributions of numerical data.
<code>pie()</code>	Generates circular pie charts representing proportions and percentages.
<code>table()</code>	Computes frequency counts and multi-dimensional contingency tables.
<code>cut()</code>	Divides continuous numeric vectors into discrete factor intervals/bins.
<code>barplot()</code>	Creates vertical bar charts, including grouped and stacked bar charts.
<code>strsplit()</code>	Splits character strings into substrings based on regex delimiter patterns.
<code>wordcloud()</code>	Generates word cloud visualizations where word size corresponds to frequency.
<code>brewer.pal()</code>	Provides color palettes from the <code>RColorBrewer</code> package for graphics.